A Distributed Projection-Based Algorithm with Local Estimators for Optimal Formation of Multi-robot System

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Abstract. In general, the optimal formation problem can be modeled as a standard constrained optimization problem according to the shape theory. By adding local supplementary estimators, it can be further modeled as a distributed constrained optimization problem. Then a distributed projection-based algorithm is designed for solving this problem. The aim of the algorithm is to drive a group of robots to move to the desired geometric pattern by minimizing the total travel distance of robots from the initial positions. It is worth noticing that, as long as the graph of the communication network among the robots is undirected and connected, the global convergence of the algorithm can be guaranteed. Moreover, all of the robots finally form an ideal formation in the limited space. Finally, simulation results are provided to verify the effectiveness of the proposed distributed algorithm.

Keywords: distributed optimal formation · projection operator · shape theory · multi-robot system

1 Introduction

Multi-robot systems have been intensively researched in recent years due to their wide applications, such as unmanned vehicle control, environmental monitoring and automated container transportation [1–4]. Among the various tasks of multi-robot systems, the formation problem is of critical importance because proper robot organization can dramatically improve coordination efficiency.

In addition, the formation of multi-robot systems is widely used in load transportation [5], satellite formation flying [3], and unmanned aerial vehicle formation [6], among other applications. The goal of multi-robot formation control is to direct robots to realize and maintain a predetermined geometric shape.

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Based on coordination schemes, the multi-robot formation control problem can be solved using leader-following control [7,8], behavior-based control (for example, obstacle avoidance and formation keeping) [9,10], and virtual structure control [11], where the formation is regarded as a single structure. According to the sensing capabilities of robots and interaction topology [12], formation control schemes can be categorized into the following groups: The first is the position-based approach [10], which assumes a global coordinate system and has a tight communication topology and low sensing capabilities. The second is the displacement-based approach [13], which communication topology and sensing capability are moderate and each robot can sense the positions of its neighbors. The third is the distance-based approach [13], which has a loose communication topology and strong sensing capabilities. In terms of analytical method, the formation of multi-robot system includes consensus-based [7], optimization-based [14] and model predictive control (MPC)-based methods [15].

The preceding paragraph describes formation control strategies for multi-robot systems. However, identifying the desired formation has received little attention. Usually, a concise way to describe formation is the shape theory [16], which focuses solely on the fundamental geometric connection between robots instead of considering factors like distance, size, and orientation. As described in [16], two formations have the same shape if they are in the same equivalence class. How to choose one formation from an equivalence class is the crux of the matter when trying to determine a desired formation. In the groundbreaking work [17], finding the optimal formation is modeled as a standard constrained optimization problem, which can be solved using various numerical algorithms. Similarly, the optimal formation problem is reformulated as a nonsmooth convex optimization in [14], and it is resolved using a recurrent neural network. In [18], a finite-time convergence algorithm is proposed for solving the formation problem with limitations on convex regions. In [19], the optimal formation problem, which is converted to a mixed norm constrained optimization problem, is solved by a projection-based iteration algorithm. However, they are all not modeled in a distributed way, which has the advantage of rapid convergence and low communication consumption.

So, we consider the distributed optimal formation of multi-robot systems in this paper. First, by introducing some auxiliary variables to estimate the states of the first two robots, the problem of determining the optimal formation is modeled as a distributed optimization problem, while many existing works for the optimal formation are centralized [14,19]. Then, we propose the projection-based iteration formulas for resolving the distributed optimization problem based on the distributed method, which greatly reduces communication consumption between robots. Finally, an illustrative example is supplied to verify the effectiveness of the proposed distributed method.

2 Problem Formulation for Optimal Formation

We are concerned with the optimal formation problem of a multi-robot system with m robots on a two-dimensional plane to ensure that robots constitute a formation as same as a given geometric shape, where the icon representing the shape (formation) consists of m points $\{s_1, s_2, \ldots, s_m\}$, denoted by S. Without loss of generality, let s_1 always lie at the origin and s_2 always be on the x axis. The multi-robot system starts with the formation $(\xi_1, \xi_2, \ldots, \xi_m) \in \mathbb{R}^{2m}$, where ξ_i is the initial position of robot i. To minimize the overall moving distance, the optimal formation problem is typically formulated as a constrained optimization problem as follows [19]:

$$\begin{aligned} \min_{P} \quad & \tfrac{1}{2}\textstyle\sum_{i=1}^{m} \|p_i - \xi_i\|_2^2, \\ \text{s.t.} \quad & P \in [S], \\ & p_i \in \Omega_i, \quad i = 1, 2, \ldots, m, \end{aligned} \tag{1}$$

in which p_i is the i-th robot's final location, $P = (p_1, p_2, \ldots, p_m)$ is the final formation and $\|\cdot\|_2$ denotes the 2-norm of vector. We denote $[S] = \{rRS + \mathbf{1}_m d^T : r \in \mathbb{R}_+, R \in SO(2), d \in \mathbb{R}^2\}$ as equivalence class of formations [16], which means that the elements of $[S]$ are generated under rotation, scale, and translation transformations from S. $\Omega_i \in \mathbb{R}^2$ is a non-empty and closed convex set that limits the moving range of robot i.

To efficiently deal with the constraint $P \in [S]$ in (1), we convert it into a series of linear equality constraints [16]:

$$M(p_i - p_1) = M_i(p_2 - p_1) \quad (i = 3, 4, \ldots, m), \tag{2}$$

where $M = \begin{pmatrix} \|s_2\|_2 & 0 \\ 0 & \|s_2\|_2 \end{pmatrix}$, $M_i = \begin{pmatrix} s_i^x & -s_i^y \\ s_i^y & s_i^x \end{pmatrix}$, $s_i = (s_i^x, s_i^y)^T$. From (2), we can observe that robot i need know the following information: p_i, p_1, p_2, M_i, and M. In other words, robots 1 and 2 have to communicate with every other robot, thus incurring significant communication consumption, as shown in Fig. (1a).

In order to counteract this drawback, the local estimators are introduced to allow each robot i $(i = 3, 4, ..., m)$ estimate the values of p_1 and p_2. In this way, robots 1 and 2 do not have to communicate with every other robot, as shown in Fig. (1b).

Assume that the i-th robot's estimates to the first and second robots' positions are p_{i1} and p_{i2} respectively. Then, the condition (2) can be transformed into

$$M(p_i - p_{i1}) = M_i(p_{i2} - p_{i1}) \quad (i = 3, 4, \ldots, m), \tag{3}$$

that is,

$$\begin{pmatrix} M & M_i - M & -M_i \end{pmatrix} \begin{pmatrix} p_i \\ p_{i1} \\ p_{i2} \end{pmatrix} = 0 \quad (i = 3, 4, \ldots, m). \tag{4}$$

Let $B_i = \begin{pmatrix} M & M_i - M & -M_i \end{pmatrix}$, $\tilde{p}_i = (p_i^T, p_{i1}^T, p_{i2}^T)^T$, then (4) can be expressed as

$$B_i \tilde{p}_i = 0 \quad (i = 3, 4, \ldots, m). \tag{5}$$

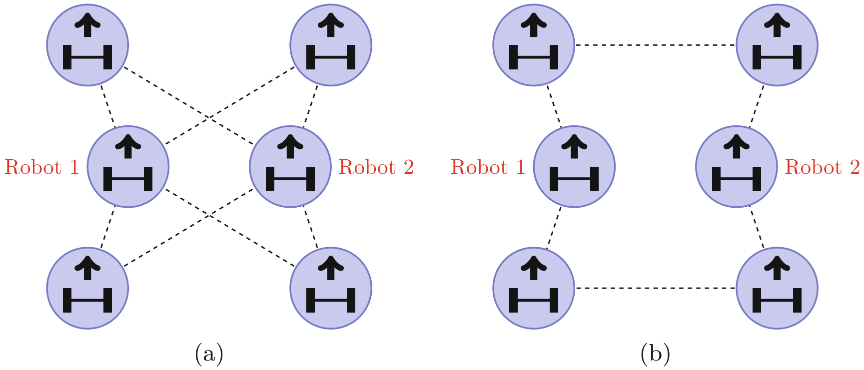

Fig. 1. (a): Robots 1 and 2 must communicate with every other robot in (2). (b): With local estimators, robots 1 and 2 only need to interact with at least one robot respectively, greatly reducing the communication load of multi-robot system.

Due to the fact that all robots have the same p_1 and p_2, the following conditions for consistence are added:

$$\begin{cases} p_{31} = p_{41} = \cdots = p_{m1} = p_1, \\ p_{32} = p_{42} = \cdots = p_{m2} = p_2. \end{cases} \tag{6}$$

We plug in the first robot's estimated value for p_2 and the second robot's estimated value for p_1 in order to design the distributed method while considering the network topology as a whole. Consequently, (6) is further expressed as follows:

$$\begin{cases} p_1 = p_{21} = p_{31} = p_{41} = \cdots = p_{m1}, \\ p_{12} = p_2 = p_{32} = p_{42} = \cdots = p_{m2}. \end{cases} \tag{7}$$

The compact form of (7) is written as

$$(L \otimes I_2)p^1 = 0, \quad (L \otimes I_2)p^2 = 0, \tag{8}$$

where $p^1 = (p_1^T, p_{21}^T, \ldots, p_{m1}^T)^T$, $p^2 = (p_{12}^T, p_2^T, \ldots, p_{m2}^T)^T$, $\otimes$ is the Kronecker product, I_2 denotes two-dimensional identity matrix. In this paper, we suppose that the interaction graph G among robots is undirected and connected, so the Laplace matrix $L = \{l_{ij}\} \in \mathbb{R}^{m\times m}$ corresponding to G has the following properties: $0 = \lambda_1 < \lambda_2 \leq \cdots \leq \lambda_m$, where λ_i is the i-th smallest eigenvalue.

To keep robots from getting too close to each other during formation switching, it is necessary to constrain the relative position between robots 1 and 2. Hence, we investigate the case for adding a free anchor constraint: $p_1 - p_2 = b$. Based on the notation introduced above, we can apply constraints $p_1 - p_{12} = b$ and $p_2 - p_{21} = -b$ to robots 1 and 2, respectively.

Combining the objective function in (1), formation constraint (5), consistency constraint (8), anchor constraint and boundary constraint $p_i \in \Omega_i$, the distributed optimization problem of multi-robot formation can be expressed as

$$\begin{array}{ll} \min\limits_{p_i} & \frac{1}{2}\sum_{i=1}^{m}\|p_i-\xi_i\|_2^2, \\ \text{s.t.} & p_1-p_{12}=b, \quad p_2-p_{21}=-b, \\ & B_i\tilde{p}_i=0 \quad (i=3,4,\ldots,m), \\ & (L\otimes I_2)p^1=0, \quad (L\otimes I_2)p^2=0, \\ & p_i\in\Omega_i \quad (i=1,2,\ldots,m). \end{array} \tag{9}$$

For the convenience of theoretical analysis, we denote

$$\tilde{p}=(p_1^T,p_{12}^T,p_2^T,p_{21}^T,p_3^T,p_{31}^T,p_{32}^T,\ldots,p_m^T,p_{m1}^T,p_{m2}^T)^T,$$

$$\tilde{c}=\mathrm{diag}\{1,0,1,0,1,0,0,\ldots,1,0,0\}\otimes I_2,$$

$$\tilde{B}=\begin{pmatrix} I_2 & -I_2 & O_2 & O_2 & O_{2\times 6(m-2)} \\ O_2 & O_2 & I_2 & -I_2 & O_{2\times 6(m-2)} \\ 0_{2(m-2)\times 2} & 0_{2(m-2)\times 2} & 0_{2(m-2)\times 2} & 0_{2(m-2)\times 2} & \mathrm{blkdiag}\{B_3,\ldots,B_m\} \end{pmatrix},$$

$$\tilde{b}=(b^T,-b^T,0_{2(m-2)}^T)^T,$$

$$\tilde{\xi}=(\xi_1^T,0_2^T,\xi_2^T,0_2^T,\xi_3^T,0_2^T,0_2^T,\ldots,\xi_m^T,0_2^T,0_2^T)^T,$$

$$\tilde{L}_1=\left\{\begin{array}{ccccccccccc} l_{11} & 0 & 0 & l_{12} & 0 & l_{13} & 0 & \cdots & 0 & l_{1m} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ l_{21} & 0 & 0 & l_{22} & 0 & l_{23} & 0 & \cdots & 0 & l_{2m} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ l_{31} & 0 & 0 & l_{32} & 0 & l_{33} & 0 & \cdots & 0 & l_{3m} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ l_{m1} & 0 & 0 & l_{m2} & 0 & l_{m3} & 0 & \cdots & 0 & l_{mm} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \end{array}\right\},$$

$$\tilde{L}_2=\left\{\begin{array}{ccccccccccc} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & l_{11} & l_{12} & 0 & 0 & l_{13} & 0 & \cdots & 0 & 0 & l_{1m} \\ 0 & l_{21} & l_{22} & 0 & 0 & l_{23} & 0 & \cdots & 0 & 0 & l_{2m} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & l_{31} & l_{32} & 0 & 0 & l_{33} & 0 & \cdots & 0 & 0 & l_{3m} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & l_{m1} & l_{m2} & 0 & 0 & l_{m3} & 0 & \cdots & 0 & 0 & l_{mm} \end{array}\right\},$$

$$\tilde{\Omega}=\Omega_1\times\Omega_2\times\Omega_2\times\Omega_1\times\Omega_3\times\Omega_1\times\Omega_2\times\cdots\times\Omega_m\times\Omega_1\times\Omega_2,$$

where diag$\{\cdot\}$ denotes the diagonal matrix, blkdiag$\{\cdot\}$ denotes the block diagonal matrix and $\times$ is the Cartesian product.

Therefore, (9) can be described as follows:

$$\begin{aligned} \min_{\tilde{p}} \quad & \tfrac{1}{2}\|\tilde{c}\tilde{p}-\tilde{\xi}\|_2^2, \\ \text{s.t.} \quad & \tilde{B}\tilde{p}=\tilde{b}, \quad \tilde{p}\in\tilde{\Omega}, \\ & (\tilde{L}_1\otimes I_2)\tilde{p}=0, \quad (\tilde{L}_2\otimes I_2)\tilde{p}=0. \end{aligned} \tag{10}$$

3 Algorithm Description and Convergence Analysis

First of all, the following lemma illustrates the necessary and sufficient conditions for the optimal solutions of (10).

Lemma 1 *[20].* *$\tilde{p}^*\in\mathbb{R}^{6m-4}$ is an optimal solution to problem (10) if and only if there exist $\tilde{y}^*\in\mathbb{R}^{2m}$, $\tilde{\theta}^*\in\mathbb{R}^{6m-4}$ and $\tilde{\eta}^*\in\mathbb{R}^{6m-4}$ such that $(\tilde{p}^*,\tilde{y}^*,\tilde{\theta}^*,\tilde{\eta}^*)$ satisfies*

$$\begin{cases} \tilde{p}^*=\phi(\tilde{p}^*-\sigma(\tilde{c}^T(\tilde{c}\tilde{p}^*-\tilde{\xi})+\tilde{B}^T\tilde{y}^*+\tilde{L}_1\otimes I_2\tilde{\theta}^*+\tilde{L}_2\otimes I_2\tilde{\eta}^*)), \\ \tilde{B}\tilde{p}^*=\tilde{b}, \\ \tilde{L}_1\otimes I_2\tilde{p}^*=0, \\ \tilde{L}_2\otimes I_2\tilde{p}^*=0, \end{cases} \tag{11}$$

where $\sigma>0$ is a constant, $\phi:\mathbb{R}^{6m-4}\to\tilde{\Omega}$ is a projection operator.

3.1 Algorithm Description

We define $\tilde{p}(k)=\tilde{p}^k$, $\tilde{y}(k)=\tilde{y}^k$, $\tilde{z}(k)=\tilde{z}^k$ and $\tilde{w}(k)=\tilde{w}^k$. From (11), a distributed algorithm is designed to solve problem (10)

$$\begin{cases} \tilde{p}^{k+1}=\phi(\tilde{p}^k-\sigma(\tilde{c}^T(\tilde{c}\tilde{p}^k-\tilde{\xi})+\tilde{B}^T(\tilde{y}^k+\tilde{B}\tilde{p}^k-\tilde{b})+\tilde{z}^k+\tilde{L}_1\otimes I_2\tilde{p}^k \\ \qquad\qquad +\tilde{w}^k+\tilde{L}_2\otimes I_2\tilde{p}^k)), \\ \tilde{y}^{k+1}=\tilde{y}^k+\tilde{B}\tilde{p}^{k+1}-\tilde{b}, \\ \tilde{z}^{k+1}=\tilde{z}^k+\tilde{L}_1\otimes I_2\tilde{p}^{k+1}, \\ \tilde{w}^{k+1}=\tilde{w}^k+\tilde{L}_2\otimes I_2\tilde{p}^{k+1}, \end{cases} \tag{12}$$

where $\tilde{z}=\left(z_1^T,z_{12}^T,z_2^T,z_{21}^T,z_3^T,z_{31}^T,z_{32}^T,\ldots,z_m^T,z_{m1}^T,z_{m2}^T\right)^T$, $\tilde{y}=\left(y_1^T,y_2^T,y_3^T,\ldots,y_m^T\right)^T$, $\tilde{w}=\left(w_1^T,w_{12}^T,w_2^T,w_{21}^T,w_3^T,w_{31}^T,w_{32}^T,\ldots,w_m^T,w_{m1}^T,w_{m2}^T\right)^T$.

It stands to reason that $(\tilde{p}^*,\tilde{y}^*,\tilde{z}^*,\tilde{w}^*)$ satisfies the equations in (11) if and only if it is also an equilibrium point of system (12) where $\tilde{z}^*=\tilde{L}_1\otimes I_2\tilde{\theta}^*$ and $\tilde{w}^*=\tilde{L}_2\otimes I_2\tilde{\eta}^*$.

Next, we present the Algorithm (12) in component form in order to fully understand the algorithm's distributed processing mechanism.

For robot 1, the iterative formula is

$$\begin{cases} p_1^{k+1} = \phi_1(p_1^k - \sigma(p_1^k - \xi_1 + y_1^k + p_1^k - b - p_{12}^k + z_1^k \\ \qquad\qquad + \sum_{j \in N_1} a_{1j}(p_1^k - p_{j1}^k))), \\ p_{12}^{k+1} = \phi_2(p_{12}^k - \sigma(-y_1^k - p_1^k + p_{12}^k + b_3 + w_{12}^k \\ \qquad\qquad + \sum_{j \in N_1} a_{1j}(p_{12}^k - p_{j2}^k))), \\ y_1^{k+1} = y_1^k + p_1^{k+1} - p_{12}^{k+1} - b, \\ z_1^{k+1} = z_1^k + \sum_{j \in N_1} a_{1j}(p_1^{k+1} - p_{j1}^{k+1}), \\ w_{12}^{k+1} = w_{12}^k + \sum_{j \in N_1} a_{1j}(p_{12}^{k+1} - p_{j2}^{k+1}), \end{cases} \tag{13}$$

where $p_{22} = p_2$ if $a_{12} \neq 0$, N_1 denotes the set of robots that can communicate with robot 1.

For robot 2, the iterative formula is

$$\begin{cases} p_2^{k+1} = \phi_2(p_2^k - \sigma(p_2^k - \xi_2 + y_2^k + p_2^k + b - p_{21}^k + w_2^k + \sum_{j \in N_2} a_{2j}(p_2^k - p_{j2}^k))), \\ p_{21}^{k+1} = \phi_1(p_{21}^k - \sigma(-y_2^k - p_2^k + p_{21}^k - b + z_{21}^k + \sum_{j \in N_2} a_{2j}(p_{21}^k - p_{j1}^k))), \\ y_2^{k+1} = y_2^k + p_2^{k+1} - p_{21}^{k+1} + b, \\ z_{21}^{k+1} = z_{21}^k + \sum_{j \in N_2} a_{2j}(p_{21}^{k+1} - p_{j1}^{k+1}), \\ w_2^{k+1} = w_2^k + \sum_{j \in N_2} a_{2j}(p_2^{k+1} - p_{j2}^{k+1}), \end{cases} \tag{14}$$

where $p_{11} = p_1$ if $a_{21} \neq 0$, N_2 denotes all neighbor robots of robot 2.

For robot i ($i = 3, 4, \ldots, m$), the iterative formula is

$$\begin{cases} p_i(k+1) = \phi_i(p_i^k - \sigma(p_i^k - \xi_i + M(y_i^k + B_i\tilde{p}_i^k))), \\ p_{i1}(k+1) = \phi_1(p_{i1}^k - \sigma((M_i^T - M)(y_i^k + B_i\tilde{p}_i^k) + z_{i1}^k + \sum_{j \in N_i} a_{ij}(p_{i1}^k - p_{j1}^k))), \\ p_{i2}(k+1) = \phi_2(p_{i2}^k - \sigma(-M_i^T(y_i^k + B_i\tilde{p}_i^k) + w_{i2}^k + \sum_{j \in N_i} a_{ij}(p_{i2}^k - p_{j2}^k))), \\ y_i(k+1) = y_i^k + B_i\tilde{p}_i^{k+1}, \\ z_{i1}(k+1) = z_{i1}^k + \sum_{j \in N_i} a_{ij}(p_{i1}^k - p_{j1}^k), \\ w_{i2}(k+1) = w_{i2}^k + \sum_{j \in N_i} a_{ij}(p_{i2}^k - p_{j2}^k), \end{cases} \tag{15}$$

where $p_{11} = p_1$ if $a_{i1} \neq 0$, $p_{22} = p_2$ if $a_{i2} \neq 0$, $\tilde{p}_i = (p_i^T, p_{i1}^T, p_{i2}^T)^T$. N_i denotes all neighbor robots of robot i.

3.2 Convergence Analysis

For further analysis, another equivalent algorithm is introduced as follows:

$$\begin{cases} \tilde{p}^{k+1} = \phi(\tilde{p}^k - \sigma(\tilde{c}^T(\tilde{c}\tilde{p}^k - \tilde{\xi}) + \tilde{B}^T(\tilde{y}^k + \tilde{B}\tilde{p}^k - \tilde{b}) + \tilde{L}_1 \otimes I_2(\tilde{\theta}^k + \tilde{p}^k) \\ \qquad\qquad + \tilde{L}_2 \otimes I_2(\tilde{\eta}^k + \tilde{p}^k))), \\ \tilde{y}^{k+1} = \tilde{y}^k + \tilde{B}\tilde{p}^{k+1} - \tilde{b}, \\ \tilde{\theta}^{k+1} = \tilde{\theta}^k + \tilde{p}^{k+1}, \\ \tilde{\eta}^{k+1} = \tilde{\eta}^k + \tilde{p}^{k+1}. \end{cases} \tag{16}$$

Theorem 1. *The algorithms in* (12) *and* (16) *are equivalent if the initial conditions meet* $\tilde{z}_0 = (\tilde{L}_1 \times I_2)\tilde{\theta}_0$ *and* $\tilde{w}_0 = (\tilde{L}_2 \times I_2)\tilde{\eta}_0$ *for any* $\tilde{\theta}_0 \in \mathbb{R}^{6m-4}$ *and* $\tilde{\eta}_0 \in \mathbb{R}^{6m-4}$.

Proof. The proof is similar to that of Lemma 3 in [20] and is omitted here. □

Next, we prove the convergence of Algorithm (16), which also holds for Algorithm (12) as per Theorem 1.

Lemma 2 [21]**.** *Let* A *and* B *are square matrices of* m *and* n *dimensions respectively,* $\lambda_1, \lambda_2, \ldots, \lambda_m$ *and* $\mu_1, \mu_2, \ldots, \mu_n$ *are eigenvalues of* A *and* B *respectively. Then* $\lambda_i\mu_j$ $(i = 1, 2, \ldots, m, j = 1, 2, \ldots, n)$ *are eigenvalues of* $A \otimes B$.

Assume that $\tilde{p}^*$ is an optimal solution to (10). From Lemma 1, there exist $\tilde{y}^*$, $\tilde{\theta}^*$and $\tilde{\eta}^*$ such that the equations in (11) hold, thus we can define and draw some conclusions about the following functions:

$$V_1(\tilde{p}^k) = \|\tilde{p}^k - \tilde{p}^*\|_2^2,\ V_2(\tilde{y}^k) = \|\tilde{y}^k - \tilde{y}^*\|_2^2,$$
$$V_3(\tilde{\theta}^k) = (\tilde{\theta}^k - \tilde{\theta}^*)^T\tilde{L}_1 \otimes I_2(\tilde{\theta}^k - \tilde{\theta}^*), V_4(\tilde{\eta}^k) = (\tilde{\eta}^k - \tilde{\eta}^*)^T\tilde{L}_2 \otimes I_2(\tilde{\eta}^k - \tilde{\eta}^*).$$

Lemma 3 [20]**.** *The following inequalities are satisfied:*

(1) $V_1(\tilde{p}^{k+1}) - V_1(\tilde{p}^k) \leq -\|\tilde{p}^{k+1} - \tilde{p}^k\|_2^2 + 2\sigma\tilde{c}^T\tilde{c}\|\tilde{p}^{k+1} - \tilde{p}^k\|_2^2 - 2\sigma(\tilde{B}\tilde{p}^{k+1} - \tilde{b})^T(\tilde{y}^k - \tilde{y}^* + \tilde{B}\tilde{p}^k - \tilde{b}) - 2\sigma(\tilde{L}_1 \otimes I_2\tilde{p}^{k+1})^T(\tilde{\theta}^k - \tilde{\theta}^* + \tilde{p}^k) - \sigma(\tilde{L}_2 \otimes I_2\tilde{p}^{k+1})^T(\tilde{\eta}^k - \tilde{\eta}^* + \tilde{p}^k)$;
(2) $V_2(\tilde{y}^{k+1}) - V_2(\tilde{y}^k) \leq 2(\tilde{B}\tilde{p}^{k+1} - \tilde{b})^T(\tilde{y}^k - \tilde{y}^* + \tilde{B}\tilde{p}^k - \tilde{b}) + \|\tilde{B}(\tilde{p}^{k+1} - \tilde{p}^k)\|^2 - \|\tilde{B}\tilde{p}^k - \tilde{b}\|^2$;
(3) $V_3(\tilde{\theta}^{k+1}) - V_3(\tilde{\theta}^k) = 2(\tilde{L}_1 \otimes I_2\tilde{p}^{k+1})^T(\tilde{\theta}^k - \tilde{\theta}^* + \tilde{p}^k) + (\tilde{p}^{k+1} - \tilde{p}^k)^T\tilde{L}_1 \otimes I_2(\tilde{p}^{k+1} - \tilde{p}^k) - (\tilde{p}^k)^T\tilde{L}_1 \otimes I_2\tilde{p}^k$;
(4) $V_4(\tilde{\eta}^{k+1}) - V_4(\tilde{\eta}^k) = 2(\tilde{L}_2 \otimes I_2\tilde{p}^{k+1})^T(\tilde{\eta}^k - \tilde{\eta}^* + \tilde{p}^k) + (\tilde{p}^{k+1} - \tilde{p}^k)^T\tilde{L}_2 \otimes I_2(\tilde{p}^{k+1} - \tilde{p}^k) - (\tilde{p}^k)^T\tilde{L}_2 \otimes I_2\tilde{p}^k$.

Theorem 2. *For any initial value* $(\tilde{p}_0, \tilde{y}_0, \tilde{\theta}_0, \tilde{\eta}_0)$*, the state* $\tilde{p}_k$ *in* (16) *is globally convergent to the optimal solution of problem* (10) *if*

$$\sigma \leq \frac{1}{\lambda_{\max}(2\tilde{c}^T\tilde{c} + \tilde{B}^T\tilde{B} + \tilde{L}_1 \otimes I_2 + \tilde{L}_2 \otimes I_2))},$$

where $\lambda_{\max}$ *is the maximum eigenvalue of corresponding matrix.*

Proof. Suppose that $\tilde{p}^*$ to be an optimal solution of (10). By Lemma 1, there exist $\tilde{y}^*$, $\tilde{\theta}^*$ and $\tilde{\eta}^*$ such that the equalities in (11) are fulfilled.

Define the candidate Lyapunov function as:

$$V(\tilde{p}^k,\tilde{y}^k,\tilde{\theta}^k,\tilde{\eta}^k)=V_1(\tilde{p}^k)+\sigma\big(V_2(\tilde{y}^k)+V_3(\tilde{\theta}^k)+V_4(\tilde{\eta}^k)\big). \tag{17}$$

Then according to Lemma 3, we have

$$\begin{aligned}
&V(\tilde{p}^{k+1},\tilde{y}^{k+1},\tilde{\theta}^{k+1},\tilde{\eta}^{k+1})-V(\tilde{p}^k,\tilde{y}^k,\tilde{\theta}^k,\tilde{\eta}_k)\\
=&V_1(\tilde{p}^{k+1})-V_1(\tilde{p}^k)+\sigma\big(V_2(\tilde{y}^{k+1})-V_2(\tilde{y}^k)\big)+\sigma\big(V_3(\tilde{\theta}^{k+1})-V_3(\tilde{\theta}^k)\big)\\
&+\sigma\big(V_4(\tilde{\eta}^{k+1})-V_4(\tilde{\eta}^k)\big)\\
\le&-\|\tilde{p}^{k+1}-\tilde{p}^k\|_2^2+2\sigma\tilde{c}^T\tilde{c}\|\tilde{p}^{k+1}-\tilde{p}^k\|_2^2-2\sigma(\tilde{B}\tilde{p}^{k+1}-\tilde{b})^T(\tilde{y}^k-\tilde{y}^*+\tilde{B}\tilde{p}^k-\tilde{b})\\
&-2\sigma(\tilde{L}_1\otimes I_2\tilde{p}^{k+1})^T(\tilde{\theta}^k-\tilde{\theta}^*+\tilde{p}^k)-2\sigma(\tilde{L}_2\otimes I_2\tilde{p}^{k+1})^T(\tilde{\eta}^k-\tilde{\eta}^*+\tilde{p}^k)\\
&+2\sigma(\tilde{B}\tilde{p}^{k+1}-\tilde{b})^T(\tilde{y}^k-\tilde{y}^*+\tilde{B}\tilde{p}^k-\tilde{b})+\sigma\|\tilde{B}(\tilde{p}^{k+1}-\tilde{p}^k)\|^2-\sigma\|\tilde{B}\tilde{p}^k-\tilde{b}\|^2\\
&+2\sigma(\tilde{L}_1\otimes I_2\tilde{p}^{k+1})^T(\tilde{\theta}^k-\tilde{\theta}^*+\tilde{p}^k)+\sigma(\tilde{p}^{k+1}-\tilde{p}_k)^T\tilde{L}_1\otimes I_2(\tilde{p}^{k+1}-\tilde{p}^k)\\
&-\sigma(\tilde{p}^k)^T\tilde{L}_1\otimes I_2\tilde{p}^k+2\sigma(\tilde{L}_2\otimes I_2\tilde{p}^{k+1})^T(\tilde{\eta}^k-\tilde{\eta}^*+\tilde{p}^k)\\
&+\sigma(\tilde{p}^{k+1}-\tilde{p}^k)^T\tilde{L}_2\otimes I_2(\tilde{p}^{k+1}-\tilde{p}^k)-\sigma(\tilde{p}^k)^T\tilde{L}_2\otimes I_2\tilde{p}^k\\
\le&-\|\tilde{p}^{k+1}-\tilde{p}^k\|_2^2+2\sigma\tilde{c}^T\tilde{c}\|\tilde{p}^{k+1}-\tilde{p}^k\|_2^2+\sigma\|\tilde{B}(\tilde{p}^{k+1}-\tilde{p}^k)\|^2-\sigma\|\tilde{B}\tilde{p}^k-\tilde{b}\|^2\\
&+\sigma(\tilde{p}^{k+1}-\tilde{p}^k)^T\tilde{L}_1\otimes I_2(\tilde{p}^{k+1}-\tilde{p}^k)-\sigma(\tilde{p}^k)^T\tilde{L}_1\otimes I_2\tilde{p}^k\\
&+\sigma(\tilde{p}^{k+1}-\tilde{p}^k)^T\tilde{L}_2\otimes I_2(\tilde{p}^{k+1}-\tilde{p}^k)-\sigma(\tilde{p}^k)^T\tilde{L}_2\otimes I_2\tilde{p}^k\\
=&-(\tilde{p}^{k+1}-\tilde{p}^k)^T\big(I-\sigma(2\tilde{c}^T\tilde{c}+\tilde{B}^T\tilde{B}+\tilde{L}_1\otimes I_2+\tilde{L}_2\otimes I_2)\big)(\tilde{p}^{k+1}-\tilde{p}^k)\\
&-\sigma\|\tilde{B}\tilde{p}^k-\tilde{b}\|^2-\sigma(\tilde{p}^k)^T\tilde{L}_1\otimes I_2\tilde{p}^k-\sigma(\tilde{p}^k)^T\tilde{L}_2\otimes I_2\tilde{p}^k.
\end{aligned}$$

If $\sigma\le 1/\lambda_{max}(2\tilde{c}^T\tilde{c}+\tilde{B}^T\tilde{B}+\tilde{L}_1\otimes I_2+\tilde{L}_2\otimes I_2)$, the matrix $I-\sigma(2\tilde{c}^T\tilde{c}+\tilde{B}^T\tilde{B}+\tilde{L}_1\otimes I_2+\tilde{L}_2\otimes I_2)$ is positive definite. And according to the definition of $\tilde{L}_1$ and $\tilde{L}_2$ as well as Lemma 2, we have $\tilde{L}_1\otimes I_2$ and $\tilde{L}_2\otimes I_2$ are positive semi-definite. Consequently, $V(\tilde{p}_{k+1},\tilde{y}_{k+1},\tilde{\theta}_{k+1},\tilde{\eta}_{k+1})-V(\tilde{p}_k,\tilde{y}_k,\tilde{\theta}_k,\tilde{\eta}_k)\le 0$.

The rest of proof is similar to that of Theorem 2 in [20], so we leave it out here due to page limit. □

4 Illustrative Example

To verify the proposed distributed algorithm, we show the simulation results of the distributed optimal formation control schemes applied to 33 robots working in a two-dimensional environment. The robot's position is represented by its coordinates. The task is to find a new formation like Fig. 2 based on the principle of minimum total travel distance. Figure 3 shows the preliminary assembly of the multi-robot system. The optimal formation problem in this paper is formulated as (10) in a distributed way. And the communication graph among robots has a ring structure.

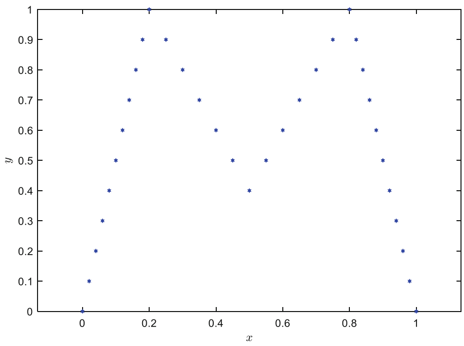

Fig. 2. Reference shape icon M.

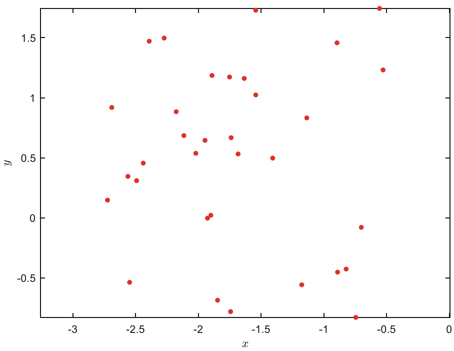

Fig. 3. The preliminary assembly of the multi-robot system.

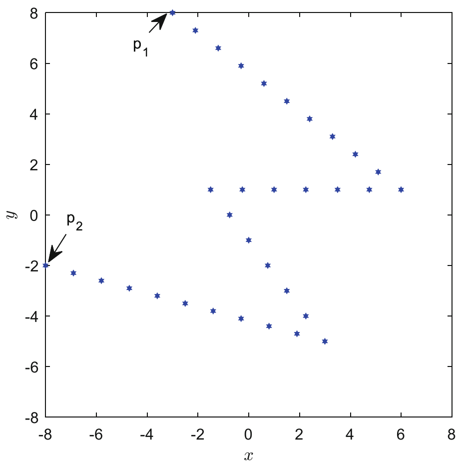

Fig. 4. Optimal formation obtained by using the Algorithm (16) with the limited region $\Omega = [-8, 8]^{2m}$.

The limited area Ω_i of robot i ($i = 1, 2, ..., 33$) is set as $[-8, 8]^2$. The optimal formation obtained by using the Algorithm (16) is shown in Fig. 4. In addition, Fig. 4 depicts the simulation formation on an example with two anchors p_1 and p_2 satisfying $p_1 - p_2 = (5, 10)^T$, indicating that the shape rotates along the direction $(5, 10)^T$.

5 Conclusion

According to the shape theory and utilizing auxiliary estimators, the optimal formation of multi-robot system has been formulated as a distributed optimization problem in this paper. Then a distributed projection-based algorithm has been presented to deal with this problem. The states of all robots obtained by the proposed distributed algorithm are convergent globally under the assumption that the communication graph is undirect and connected. Moreover, all robots have formed an ideal formation within the limited area. At the end, simulation results are given to show the effectiveness of the designed distributed technique.

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